25900692 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Core Course (DSC) - MG3DSCZIM201 - BACTERIAL PHYSIOLOGY AND METABOLISM

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Fill in the blanks

Answer all questions

Each question carries **1** marks

1. …..... use light as their energy source [K] / [CO1]
2. Vancomycin inhibits bacterial …..... synthesis [K] / [CO4]
3. Anaerobic respiration produces less …..... compared to aerobic respiration [K] / [CO5]
4. …..... is mainly obtained by plants from the atmosphere in the form of carbon dioxide [K] / [CO1]
5. UV radiation causes …..... damage by forming thymine dimers [U] / [CO3]
6. Differential media allows microorganisms to be differentiated based on their…..... characteristic such as sugar fermentation or enzyme production. [K] / [CO2]
7. Continous culture are useful in studying…..... under controlled conditions [K] / [CO3]
8. In oxidative phosphorylation ATP is produced by the …..... [U] / [CO5]

9. Glycolysis is a metabolic pathway that converts ….... into pyruvate [K] / [CO5]

10. The dilution method using broth in microplates is called the …..... method [U] / [CO4]

[1x10 = 10]

Part B

Short Answer Questions

Answer any **10** questions

Each question carries **2** marks

11. Explain the role of phosphoenolpyruvate in the phosphotransferase system. [K] / [CO1]

12. What are the medical applications of silver compounds? [U] / [CO3]

13. What is the biological significance of the ED pathway? [U] / [CO5]

14. Explain the mechanism of action of beta-lactam antibiotics. [U] / [CO4]

15. What is lactic acid fermentation ? Give examples. [K] / [CO5]

16. What is the benefits of using the spread plate technique? [K] / [CO2]

17. How does a turbidostat differ from a chemostat? [K] / [CO3]

18. What is the primary function of the TCA Cycle? [U] / [CO5]

19. Define active transport. [K] / [CO1]

20. How does coenzyme Q facilitate electron transfer in the ETS process? [U] / [CO5]

21. Why are cultural characteristics important in research and quality control? [U] / [CO2]

22. What are the limitations of using low temperature as a microbial control method? [K] / [CO3]

[2x10 = 20]

Part C

Short Essay Questions

Answer any **5** questions

Each question carries **4** marks

23. Describe the regulation of the HMP pathway and the role of glucose -6-phosphate dehydrogenase deficiency . [U] / [CO5]

24.	Describe the importance of selecting the appropriate media components for the growth and isolation of specific microorganism.	[U] / [CO2]
25.	Explain the viable plate count method with steps and its significance in microbiology.	[K] / [CO3]
26.	Explain the chemical types of media	[U] / [CO2]
27.	Discuss the factors that contribute to the onset of the stationary phase in a growth curve.	[U] / [CO3]
28.	Describe cryopreservation and its advantages in microbial preservation.	[U] / [CO2]
29.	Discuss the involvement of ATP in biosynthetic reaction.	[K] / [CO5]

[4x5 = 20]